

Workshop Script

Interactive Museum Display & Data Exploration

Welcome

Facilitator:

"Welcome, everyone! Thank you for joining today's workshop on interactive museum displays and data visualization. My name is [Your Name] and I am working as a [Your Role] at Indiana University, Indianapolis."

"On behalf of our research team working at **IES Lab which is Interactive Embodied Space Lab** - I'm excited to explore how we can design more intuitive and engaging museum experiences that can engage visitors through embodied movements and data visualization."

Recording Consent

"Before we begin, I'd like to inform you that this session will be **recorded** for research purposes. The recording will help us analyze patterns and insights from this discussion, ensuring that your valuable input contributes to improving interactive museum experiences."

"We assure you that the recording will be used **only for research purposes** and will not be shared publicly. Your privacy is important to us."

"If you are comfortable with this, we kindly ask for your permission to record the session. If anyone has concerns or prefers not to be recorded, please let us know now."

(Wait for everyone's response)

"Thank you all for your coordination, Let's start the workshop!"

Introduction

Today museums are evolving into dynamic spaces where technology and interaction enhance learning. And we wanted to understand how **embodied movements**—the way we use our bodies to interact with digital systems—can enhance data visualization and learning in museum environments.

The goal of this workshop is to explore how interactive museum displays use movement-based interactions, how they impact data visualization, and how we can refine these experiences for better user engagement.”

“Before we begin, let’s do a quick round of introductions. Please share your name and the organization you’re with. Feel free to also mention your experience with interactive exhibits or data visualization!”

“Okay, Let me start with my experience in this space, My name is [Your Name], I’m working with IU in one of it’s research lab as a [Your Role] to [Talk about your work].”

“That’s all about us! Now who’s next?” (Wait for someone to volunteer to reply)

Start here- (Introduction Video)

“Before we begin the activities, we want to give you some background on the project and the key concepts we’ll be exploring today.”

"First, you’ll see a short video explaining an **interactive display** we developed.

"After watching, feel free to ask any questions!"

(Play the **Interactive Display Video**.)

"Do you have any questions about what you just saw? Feel free to unmute or drop them in the chat."

(Pause for questions and address any clarifications.)

Clarification & Disclaimer:

"Before we dive in, I want to clarify that today's session is **not** about critiquing a specific prototype shown in the previous video but rather gathering your insights about embodied interaction within exhibits.”

(CMT Concept and Video)

"Next, we’ll introduce a key concept that shapes our research: **Conceptual Metaphor Theory (CMT)**."

"In simple terms, CMT explains how our bodily experiences influence the way we understand abstract concepts. For example, we often associate 'up' with 'more'—like graphs showing an upward trend for increases."

"Let's watch a short video to understand this better."

(Play the CMT Video.)

"Any thoughts or questions before we move on to the activities?" (Pause briefly for engagement.)

"Now that we have this foundation, let's dive into several hands-on activities, using sticky notes and discussions to explore these concepts. I will provide clear instructions, examples, and prompts, so don't worry—just have fun and share your insights!"

Activity Overview

Facilitator:

"Let's begin with activity overview"

"This workshop is structured into five key tasks. Each activity builds on the last, helping us understand different aspects of interactive museum displays:"

✅ **Task 1: Exploring Functionality** where we will identify common gestures and movements in interactive museum spaces.

✅ **Task 2: Data Visualization Functionality** where we will understand how movements influence data representation.

✅ **Task 3: Data Literacy Goals** where we will define what users learn from interactive visualizations.

✅ **Task 4: Learning Through Schema** where we will get familiar with schemas and how they shape interactions.

✅ **Task 5: Pairing & Structuring** where we will match movements, schemas, and functionalities to create a structured understanding.

"I will guide you through each task and provide examples to help frame your thinking. Let's get started!"

FigJam Link Sharing:

"How many of you are familiar with Figma?" (Wait for their Reply)

" For people who don't know – do not worry, we are here to help you !"

"We have shared a link in the chat that will take you to our **FigJam board**—a collaborative space where we can work together in real time. This is where you'll be adding your observations, insights, and notes throughout the workshop."

 Please click the link and join the board now.

"Once you're in, take a moment to familiarize yourself with the space. You can use sticky notes to add your thoughts, move elements around, and interact with others' contributions. I'll show you demo. "

Demo of FigJam :

"At the bottom of the FigJam board, you'll see the **sticky notes** tool. You can **drag and drop** a sticky note anywhere on the board. Once you place it, just click on it to start typing your thoughts."

"For now, don't worry about changing colors—just focus on adding your ideas and interacting with others' notes."

(Pause to ensure everyone joins.)

"Can everyone access the FigJam board? If you're having any trouble, please let us know in the chat, and we'll assist you."

(Once confirmed, move forward with the activity.)

Activity Instructions

Task 1: Exploring Functionality which is body movements and gestures

Objective of this task is to identify gestures and body movements commonly used in interactive museum displays.

 Here are **Instructions** :

1. **Recall past interactions in museum spaces.**

- a. *Think about interactive data visualizations you have seen. Where was it? What exhibit? How did you interact with it?*

2. List gestures used (e.g., swiping, waving, stepping, pointing).

Only when conversation does not start ----

3. Describe how you performed or observed them.

- a. *For example: If you swiped to navigate, was it left-right, or up-down? If you pointed, what were you trying to select?*

4. Share & Discuss:

- a. *How intuitive or challenging were these gestures? Did the display respond as expected?*

💡 For an **Example :** (**only if they get distracted from topic**)

"I waved my hand to reveal historical data on a timeline. When I moved my hand left, it showed earlier events, and when I moved it right, it revealed more recent events. It felt very natural because it mimicked flipping through pages of a book."

"As I showed you before, pick a sticky add it to the blue box – task 1 activity space and start writing your thoughts. If you want to verbally share it, we are open to that as well. "

(Let them discuss and share)

"Okay! We got interesting observations over here. Well done everyone! With this, we have completed our first task."

Task 2: Data Visualization Functionality

Objective of this task is to understand how gestures influence changes in data visualization.

📌 Here are **Instructions:**

1. Describe what was triggered by the movement change in the data visualization.

- Did the display change colors, shapes, or sizes?
- Did objects move, zoom, or highlight?
- What was the overall functionality? (e.g., Filtering data, zooming in, revealing layers of information).

💡 For an **Example :** (**only when conversation does not start**)

"I stepped forward in front of a data map, and as I got closer, the display zoomed in on specific

details. When I stepped back, it zoomed out to show a broader view. The data changed dynamically with my movement, which made it feel intuitive and immersive."

"Pick a sticky add it to the orange box – task 2 activity space and start writing you thoughts. Again If you want to verbally share it, we are open to that as well. "

(Let them discuss and share)

"Interesting! With this, we have completed our second task."

Task 3: Data Literacy Goals

Objective for this task is to Define what visitors should learn from the interactive visualization.

✚ Here are **Instructions**:

- 1. List the knowledge visitors gained from the display.**
 - a. Did it effectively communicate insights?
 - b. Did the interaction help users see patterns, trends, or relationships in data?
 - c. Was it easy to interpret, or did it require additional context?
- 2. What do you think people should learn from these interactions with data visualizations?**

💡 For an **Example : (only when conversation does not start)**

"In an exhibit about climate change, I used a slider by oving my hand left right to adjust CO₂ levels over time. The display responded by changing the color of a world map—darker red indicated higher temperatures. This helped me understand the direct relationship between CO₂ and global warming trends."

"Pick a sticky add it to the violet box – task 3 activity space and start writing you thoughts or share it verbally."

(Let them discuss and share)

"Thank you for sharing this insights with us ! With this, we have completed our third task."

Task 4: Learning Through Schema

Objective for this task is to familiarize yourself with the schema representation and its role in interactive displays.

✦ Here are **Instructions**:

1. Look at the list of schemas and choose 5.

- a. Use a body movement or gestures to physically enact it.

"In the green activity box, you'll see different schemas listed. Please choose **any five** that stand out to you or seem most relevant."

"Once you've selected your five schemas, we'll use them in the next steps to explore how they connect to interactive data visualizations."

"If you have any questions while selecting, feel free to ask!"

💡 For an **Example** :

"In an interactive timeline exhibit, I swiped my hand to the right, and the timeline moved forward to show future events. When I swiped left, it moved backward to display historical events. This interaction followed the '**left-to-right progression**' schema, where moving right represents moving forward in time, and moving left represents going back—just like reading a book or scrolling through a digital timeline."

"Pick a sticky add it to the green box – task 4 activity space and start writing you thoughts or share it verbally."

(Let them discuss and share)

"We appreciate your effort in showing us gestures to represent schemas. I hope you enjoyed too! With this, we have completed our forth task."

Task 5: Pairing & Structuring Interactions (15 min)

Objective for this task is to connect gestures, schemas, learning goals and functionalities.

✦ Here are **Instructions**:

1. Look back at the schemata previously discussed.

- a. How do they relate to the interaction you have seen or performed?*
- b. How do they relate to the functionalities of the data visualizations we have listed.*
- c. How do they connect to the learning goals that we have identified?*

“As you can see in this yellow activity area, we have provided a space for you to map out your insights from previous tasks. Pick your key sharing and start categorizing it into the predefined categories..”

(Let them map out)

“Here you go, thank you for paring and structuring your ideas. With this, we have completed our fifth and the last task for the workshop.”

Wrap up and closing discussion

"Great job, everyone! Before we wrap up, let's reflect on what we've learned today:"

- What insights stood out to you? (Allow them to answer)
- Did any interactions or schemas surprise you? (Allow them to answer)

"Alright, Your feedback today will help shape future museum interaction designs. Thank you for sharing your expertise and engaging in this session!"

“ We appreciate your time and your valuable insights. Hope you all had fun today .”

(Allow final thoughts and conclude the workshop.)

Welcome

This workshop is designed to be hands-on and collaborative. We'll be going through a series of activities for that.

Your insights and expertise are valuable in shaping better interactive experiences. Feel free to share your thoughts, ask questions, and have fun with the activities!

Before we begin, let's start with a quick introduction. Please share your name, background, and any experiences you have with interactive museum exhibits.

Now, let's dive in! We'll start with a short video introducing the project, followed by an overview of Conceptual Metaphor Theory, which will guide some of our discussions today."

(Proceed with intro)

1. Introduction

Facilitator:

Today, we will explore how body movements influence data visualization and how we can design more effective and intuitive interactions. We will begin with a short **introductory video** showcasing the project, followed by an overview of **Conceptual Metaphor Theory (CMT)**—which explains how our bodily experiences shape our understanding of abstract concepts. Let's watch the CMT video to set the stage."

(Play intro & CMT videos)

"Now that we have an understanding of the project and CMT, let's move into the hands-on activities where we will analyze movement, data visualization, and usability."

2. Activity Overview

Facilitator:

"This workshop is structured into five key tasks. Each activity is designed to build on the previous one, helping us understand:

1. How people interact with museum displays (Task 1)
2. How those interactions influence data visualization (Task 2)
3. What visitors should learn from these interactions (Task 3)
4. How schemas shape data understanding (Task 4)

5. How we can structure and pair interactions with data representation (Task 5)

"I will guide you through each task, and you will have time to share your thoughts along the way."

3. Activity Instructions

Task 1: Exploring Functionality

Facilitator:

"In this task, we'll focus on the body movements and gestures commonly used in interactive museum spaces.

Instructions:

- Recall past experiences with interactive museum displays.
- List the gestures you used or observed (e.g., swiping, waving, stepping, pointing).
- Describe how they were performed and whether they felt intuitive or challenging.
- Share and discuss common patterns with the group.

Let's take 5 minutes to write down our thoughts, and then we'll discuss."

(Allow time for individual reflection and discussion.)

Task 2: Functionality of Data Visualization

Facilitator:

"Now that we've discussed gestures, let's explore how these movements affect data visualization.

Instructions:

- Describe gestures that triggered changes in data displays.
- Observe how data responded visually (e.g., color shifts, shape resizing, animation changes).
- Discuss whether the feedback was clear, engaging, and easy to interpret.

Think about a time when a display reacted exactly as expected—or perhaps when it didn't! Jot down your thoughts, then we'll discuss."

(Facilitate group sharing.)

Task 3: Data Literacy Goals

Facilitator:

“Now, let’s think about what visitors should learn from interacting with these displays.

Instructions:

- Identify key takeaways museum visitors gained from interacting with the display.
- Discuss whether the current data visualization helped in understanding complex data.
- Suggest improvements to enhance clarity and engagement.

Let’s take a few minutes to reflect, then share our thoughts.”

(Lead discussion on insights and possible improvements.)

Task 4: Learning Activity – Understanding Schema

Facilitator:

“Schemas help us understand and organize information. In this task, we will explore different schemas and how they shape interactions.

Instructions:

- Look at the provided schema examples.
- Enact how you would perform them using body movements.
- Relate schema understanding to interactive data displays.

This will help us see how intuitive or complex certain movements are for users. Let’s try it out!”

(Encourage participants to physically enact and discuss their schemas.)

Task 5: Pairing & Structuring Interactions (15 min)

Facilitator:

“Finally, let’s connect everything together—gestures, schemas, and data visualization functionalities.

Instructions:

- Match different gestures or interactions to their corresponding schema and function.
- Identify how each interaction aligns with data visualization goals.
- Discuss how this mapping improves usability and learning.

4. Create an Index Card

We will use **index cards** for this exercise. Each card includes:

- ✓ **Schema** – The movement concept (from Task 4).
- ✓ **Body movement** – The gestures identified in Task 1.
- ✓ **Functionality** – How movements influence data visualization (from Task 2).
- ✓ **Data Literacy Goal** – What users should learn (from Task 3).
- ✓ **Other Considerations** – Any usability or accessibility concerns.

Let's work together to organize and match these elements."

(Guide participants as they map interactions and discuss their choices.)

5. Wrap-Up Discussion & Closing

Facilitator:

"Great job, everyone! Before we wrap up, let's reflect on what we've learned:

- What insights stood out to you?
- Did any interactions or schemas surprise you?
- How can we improve these experiences for museum visitors?

Your feedback will help refine our approach to interactive displays. Thank you all for your participation and valuable insights!"

(Facilitate final discussion, then conclude the workshop.)

